

ALGORITHM

Spring 2025, Midterm Exam, April 9

5:00 – 8:00 pm

1. (15%) Illustrate the operation of the following sorting algorithms respectively on the array $A = \langle 3, 7, 1, 4, 0, 6, 4, 2, 1, 3, 4 \rangle$, where $A[j] \in \{0, 1, \dots, 7\}$ for $1 \leq j \leq 11$.

Which of them are stable sorting algorithms? Which of them are in-place sorting algorithms?

(a) MERGE SORT

(b) QUICKSORT

(c) HEAPSORT

2. (20%) Let $f(n)$ and $g(n)$ be real-valued functions. The asymptotic notations, O , Ω , and Θ , are defined as follows.

- $f(n)$ is *asymptotically upper bounded by* $g(n)$, denoted by $f(n) = O(g(n))$,
iff there exists positive constants c and n_0 such that $f(n) \leq c \cdot g(n)$ for all $n > n_0$.
- $f(n)$ is *asymptotically lower bounded by* $g(n)$, denoted by $f(n) = \Omega(g(n))$,
iff there exists positive constants c and n_0 such that $f(n) \geq c \cdot g(n)$ for all $n > n_0$.
- $f(n)$ is *asymptotically equivalent to* $g(n)$, denoted by $f(n) = \Theta(g(n))$,
iff there exists positive constants c_1, c_2 and n_0 such that $c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$
whenever $n > n_0$.

Please answer the following questions according to the definitions given above.

(a) Finding upper bound for the function $f(n) = n^2 + 2n - 6$ by using the O notation.
Justify your answer by demonstrating the constants.

(b) Finding lower bound for the function $f(n) = n^2 + 2n - 6$ by using the Ω notation.
Justify your answer by demonstrating the constants.

(c) Finding tight bound for the function $f(n) = n^2 + 2n - 6$ by using the Θ notation.
Justify your answer by demonstrating the constants.

(d) Show that $n^2 + 2n - 6 \neq \Theta(n^4)$.

3. (15%) Use a recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 3T(\lfloor n/3 \rfloor) + 2n$, where $T(1) = 2$ and $T(2) = 6$. Use the substitution method to verify your answer.

4. (15%) Given an array $A = \langle x_1, x_2, \dots, x_n \rangle$ of numbers, the maximum-subarray problem is to find two indices i and j such that $A = \langle x_i, x_2, \dots, x_j \rangle$ has the greatest sum of any contiguous subarray of array A .
- (a) Illustrate the operation of FIND-MAXIMUM-SUBARRAY on the array $A = \langle -3, 7, -2, 4, -3, 3, 4, -7 \rangle$.
 - (b) Write pseudocode for the brute-force method of solving the maximum-subarray problem. Your procedure should run $\Theta(n^2)$ time.
5. (15%) Please answer the following questions.
- (a) Use the master method to give a tight asymptotic bound on the running time of the recurrence $T(n) = 7T(n/2) + n^3$.
 - (b) Illustrate the operation of MAX-HEAP-INSERT($A, 3$) on the heap $A = \langle 15, 13, 9, 10, 12, 8, 7, 4, 5, 6, 2, 1 \rangle$. node i from heap A . Give an implementation of HEAP-DELETE that runs in $O(\lg n)$ time for an n -element max-heap.
 - (c) Give an example to show that the worst-case running time of MAX-HEAPIFY on a heap of size n is $\Omega(\lg n)$.
6. (15%) A lazy programmer thinks that implementing the procedure MERGE (of the merge sort) is too complicated. To simplify MERGE-SORT he eliminates the merge all together, and only recursively “sorts” the two halves of the array. Here is his pseudo code.

LAZY-SORT(A, p, r)

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1  if  $p < r$ 
2       $q \leftarrow \lfloor (p + r)/2 \rfloor$ 
3      LAZY-SORT( $A, p, q$ )
4      LAZY-SORT( $A, q + 1, r$ )

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Initial call: LAZY-SORT($A, 1, n$)

- (a) What does LAZY-SORT do?.
 - (b) What is the running time of LAZY-SORT?
7. (15%) Finding a missing number. An array of n elements contains all but one of the integers from 1 to $n + 1$.
- (a) Give the best algorithm you can for determining which number is missing if the array is sorted, and analyze its asymptotic worst-case running time.
 - (b) Give the best algorithm you can for determining which number is missing if the array is “not” sorted, and analyze its asymptotic worst-case running time.